

Custom Compiler Grammar Rules

Daniel Greenfield

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1 Intro

1.1 Contents

This contains the initial structure of grammar to be used for the first iteration of my primitive custom programming language.

1.2 Notation

For this specification we will follow EBNF form therefore $[\dots]$ denotes none or once and $\{\dots\}$ denotes none or more.

2 Lexical

For this section letter will refer to any letter(upper or lower case) in the Latin alphabet and digit will refer to any Arabic numeral.

$\langle \text{keyword} \rangle \rightarrow \text{"function"} \mid \text{"return"} \mid \text{"let"} \mid \text{"while"} \mid \text{"if"} \mid \text{"else"}$
 $\langle \text{id} \rangle \rightarrow (\langle \text{letter} \rangle \mid \text{"_"}), \{ \langle \text{letter} \rangle \mid \langle \text{digit} \rangle \mid \text{"_"} \}$
 $\langle \text{intLit} \rangle \rightarrow \langle \text{digit} \rangle, \{ \langle \text{digit} \rangle \}$
 $\langle \text{floatLit} \rangle \rightarrow \langle \text{digit} \rangle, \{ \langle \text{digit} \rangle \}, \text{"."}, \langle \text{digit} \rangle, \{ \langle \text{digit} \rangle \}$
 $\langle \text{boolLit} \rangle \rightarrow \text{"true"} \mid \text{"false"}$
 $\langle \text{type} \rangle \rightarrow \text{"u32"} \mid \text{"f32"} \mid \text{"bool"}$
 $\langle \text{punctuation} \rangle \rightarrow \text{"@"} \mid \text{";" } \mid \text{"{" } \mid \text{"}" } \mid \text{"(" } \mid \text{")"}$
 $\langle \text{Operand} \rangle \rightarrow \text{"++"} \mid \text{"--"} \mid \text{" " } \mid \text{"+" } \mid \text{"-"} \mid \text{"*"} \mid \text{" /"} \mid \text{"//"} \mid \text{"%"} \mid \text{">"} \mid \text{"<"} \mid \text{"="}$
 $\mid \text{"\&"} \mid \text{"|" } \mid$